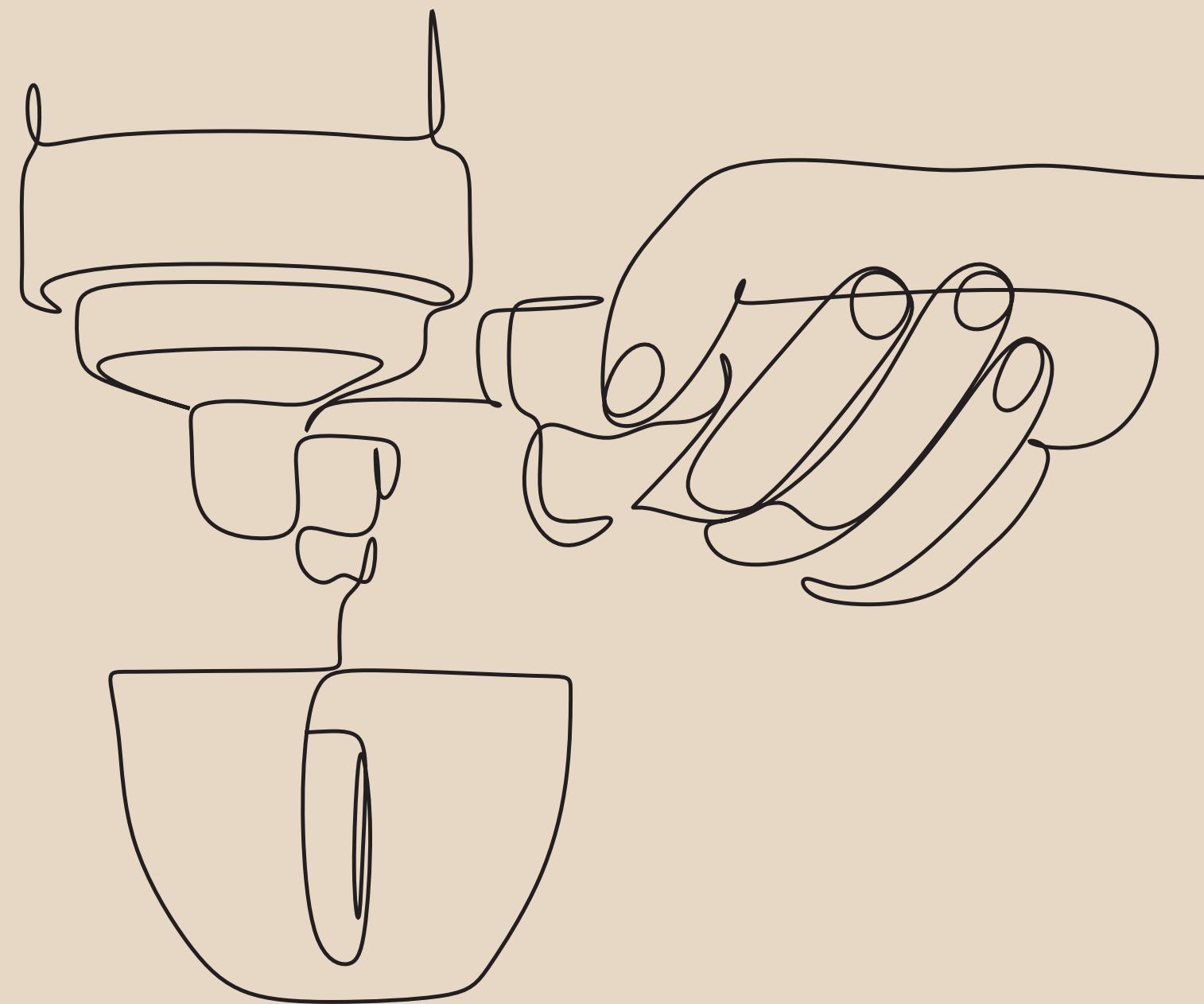


COFFEE SALES ANALYTICS

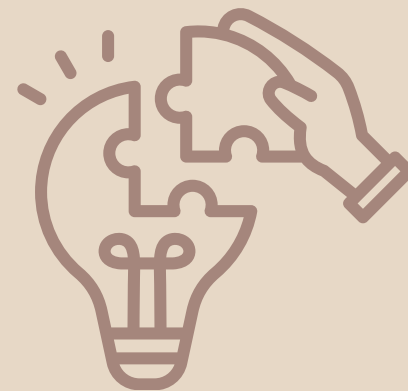
END-TO-END BUSINESS INTELLIGENCE
CASE STUDY



BUSINESS CONTEXT & CHALLENGES

Business Background

- **Dynamic Retail Growth:** The analysis covers a high-volume coffee retail location generating thousands of transactions, where rapid scale required a more mature, data-driven approach to operations.



Core Problems

- **No Single Source of Truth:** Transactional data was fragmented and unstructured, making it impossible to perform rapid, reliable health checks on financial performance.
- **Hidden Product Profitability:** Difficulty in cross-referencing sales volumes with actual revenue per menu item, caused by visual noise and extreme outliers in raw data.
- **Unknown Operational Load:** Lack of clear visibility into hourly traffic peaks and shifting consumer preferences across different service windows and calendar types (Workdays vs. Weekends).

Project Objective

- **Consolidation & Transformation:** Build a reproducible SQL ETL data pipeline to clean raw records and automate the pre-calculation of core business metrics.
- **Premium UI/UX Visualization:** Deliver a dual-page, intuitive Power BI application to empower management in optimizing menu engineering and streamlining staff scheduling.

SOLUTION ARCHITECTURE & TECH STACK

The Data Pipeline

RAW DATA
Thousands of unstructured transaction records.



ETL & CLEANING
Data consolidation, metadata mapping, and advanced feature engineering.

VALIDATION
Quality assurance audit and structured export preparation.



VISUALIZATION
DAX measures, and interactive UI/UX.

"SQL-First" Architecture

- **Engine Optimization:** Shifting logic to the database layer minimizes .pbix weight and accelerates refresh times.
- **Future-Proofing:** SQL View-based architecture ensures seamless migration to enterprise-scale systems in the future.

DATA ENGINEERING & SQL LOGIC

Key Feature Engineering

- **Time-Based Segmentation:** Transforming raw timestamps into operational windows (Morning, Evening) to identify peak flows.
- **Workday Flagging:** Using binary identifiers (0/1) to differentiate consumer behavior on workdays vs. weekends.
- **Automated Seasonality:** Derived seasons from month indices directly in SQL to track long-term trends.
- **Metadata Enrichment:** Merging raw sales with product dimensions (Caffeine level, category).

Optimization Strategy:

- **"SQL-First" Approach:** Shifting transformation logic to the database layer to ensure maximum Power BI performance.

```
CREATE VIEW v_coffee_analysis as
select
  cs.sell_date,
  cs.date_time,
  DAYNAME(cs.sell_date) AS day_name,
  WEEKDAY(cs.sell_date) AS day_index,
  CASE
    WHEN WEEKDAY(cs.sell_date) IN (5, 6) THEN 0
    ELSE 1
  END AS is_workday,
  CASE
    WHEN MONTH(cs.sell_date) IN (12, 1, 2) THEN 'Winter'
    WHEN MONTH(cs.sell_date) IN (3, 4, 5) THEN 'Spring'
    WHEN MONTH(cs.sell_date) IN (6, 7, 8) THEN 'Summer'
    ELSE 'Autumn'
  END AS year_season,
  cs.money,
  cs.coffee_name,
  cm.category,
  cm.caffeine_level,
  cm.contains_milk,
  ts.segment_name,
  ts.segment_id #
from coffee_sales cs
left join coffee_metadata cm on cs.coffee_name = cm.coffee_name
left join time_segments ts on TIME(cs.date_time) BETWEEN ts.start_time AND ts.end_time
```

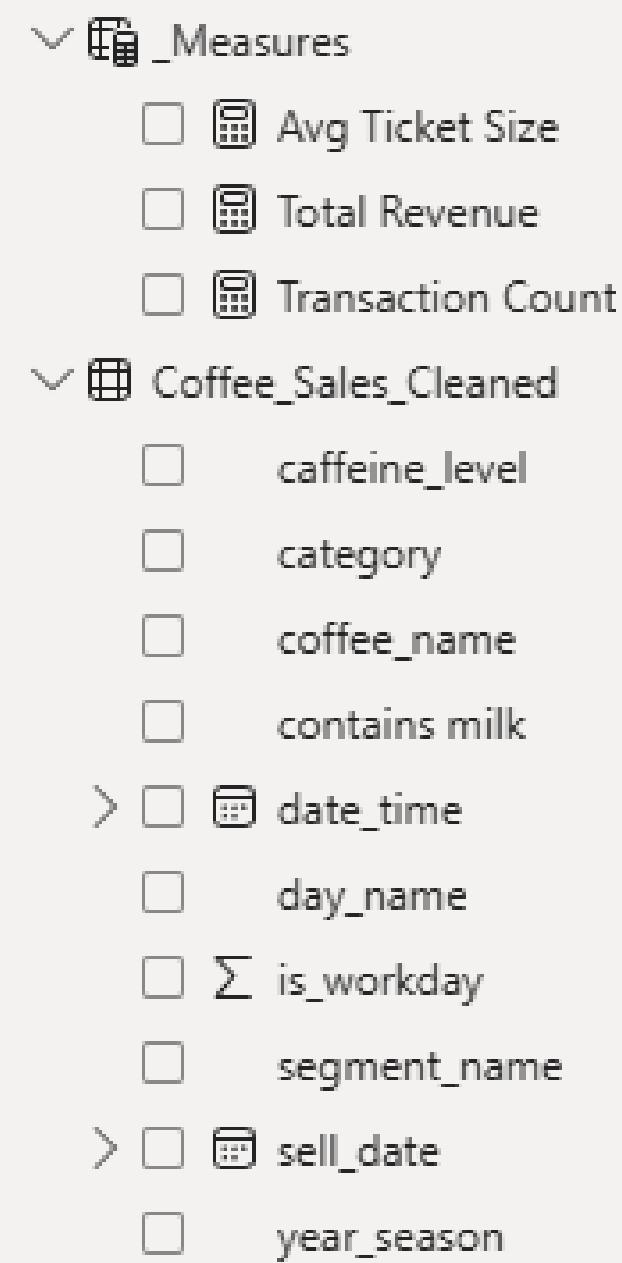
DATA MODELING & SEMANTIC LAYER GOVERNANCE

Model Architecture: Unified Flat Table

- **SQL-Optimized Structure:** For this data volume, the model utilizes a single, high-performance table fully pre-processed at the SQL View level. This architecture ensures maximum query speed and eliminates unnecessary relational complexity.

Data Governance & Best Practices

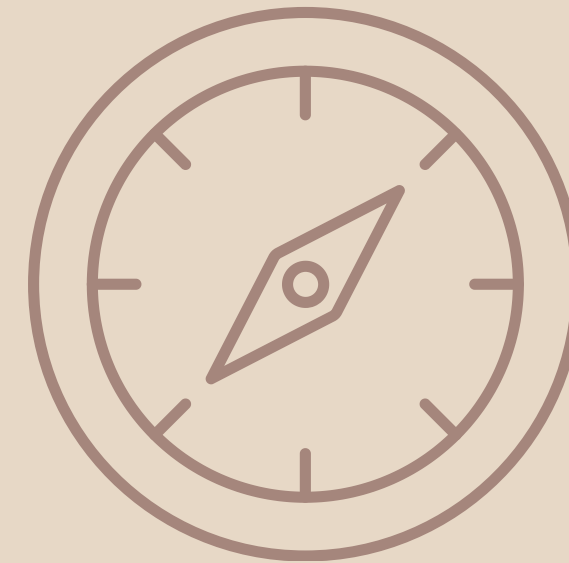
- **DAX Measure Registry:** All explicit business logic is consolidated into a dedicated _Measures table. This isolation separates calculations from the raw data layer, following professional BI documentation standards.
- **Self-Service BI Readiness:** Technical keys and raw fact columns (e.g., money, day_index, segment_id) are proactively hidden. This prevents end-user errors, such as incorrect automatic summation of ID fields.
- **Clean Presentation Layer:** The interface is stripped of technical noise, providing an intuitive field list focused on business categories and validated metrics.



INTERACTIVITY & USER EXPERIENCE

Navigation & View Management (Bookmarks)

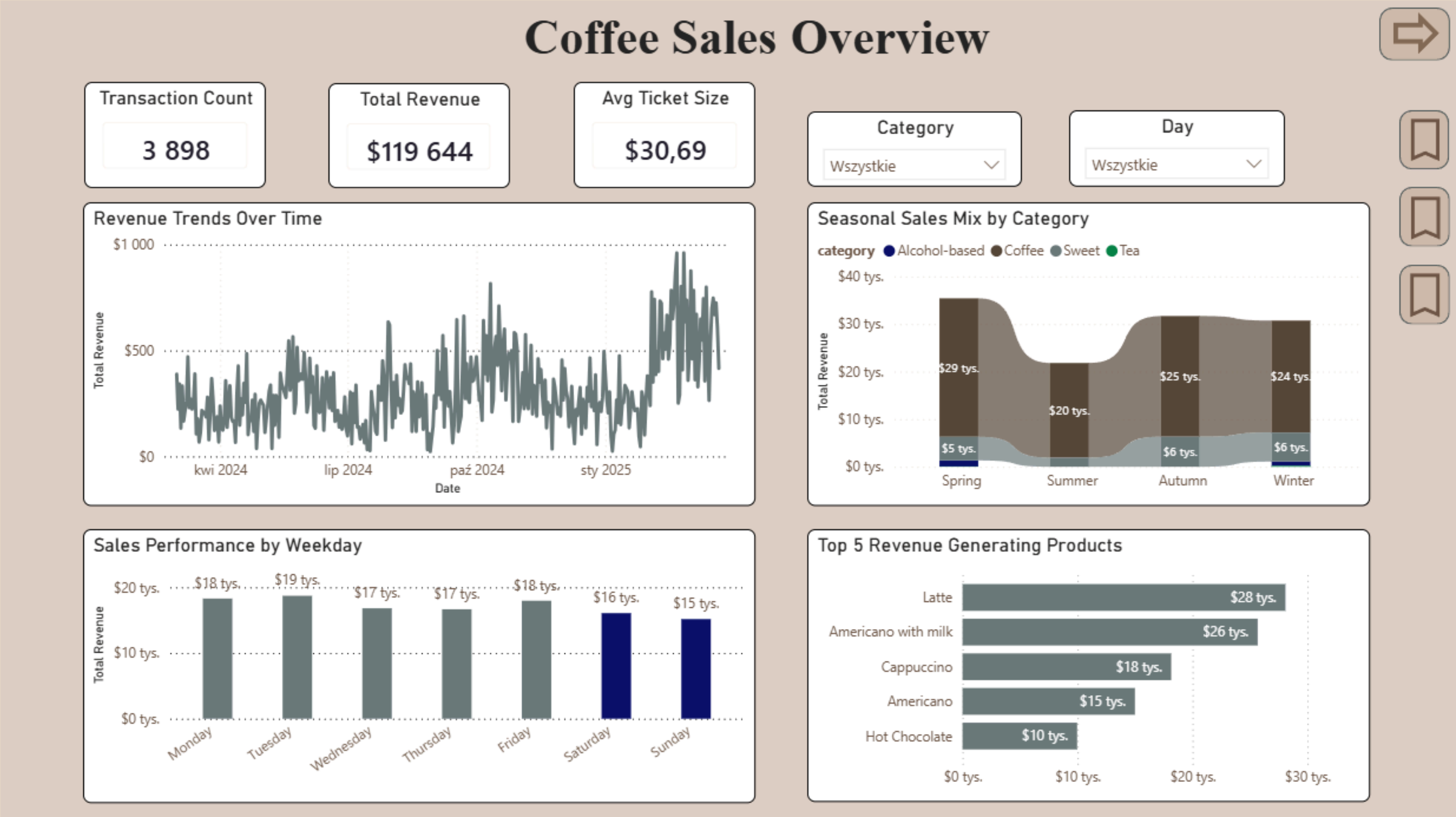
- **Intuitive Page Navigation:** Implemented dynamic navigation buttons allowing seamless switching between strategic and operational views without relying on Power BI's default bottom tabs.
- **Scenario-Based Analytics:** Engineered dedicated bookmarks (Workdays vs. Weekends) that change the entire report context with a single click, enabling users to instantly compare distinct customer behavior models.
- **User-Friendly Reset Button:** Integrated a "Clear All Filters" function, effortlessly restoring the dashboard to its initial, default state.



Features Supporting End-Users

- **Seamless Cross-Filtering:** Enabled full interactivity between visuals—selecting a specific product category or season dynamically filters all other modules, making trend discovery effortless.
- **Optimized Information Hierarchy:** Arranged the layout from macro to micro, ensuring users naturally absorb high-level KPIs before diving into continuous trend charts and granular detail tables.

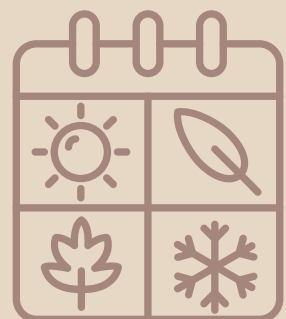
EXECUTIVE SALES INSIGHTS



EXECUTIVE SALES INSIGHTS

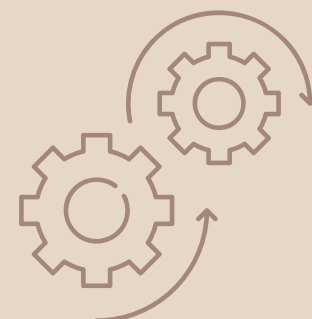
Seasonality & Menu Gap

- **Insight:** The dashboard reveals an unconventional revenue drop during the Summer season. This indicates a high probability of a menu gap, specifically a lack of refreshing, cold-beverage alternatives.
- **Recommendation:** Expand the Summer product line by introducing specialized cold brews, iced lattes, and frappes to capture warm-weather foot traffic.



Weekly Dynamics & Commuter Profiling

- **Insight:** Transaction volume strictly peaks on workdays, with a heavy concentration on Mondays and Tuesdays. This strongly correlates with corporate commuter behavior (pre-work and mid-day coffee runs).
- **Recommendation:** Dynamic staffing optimization—increase barista headcount during morning and afternoon peaks on early workdays to accelerate throughput and eliminate queue dropouts.



Product Mix Optimization

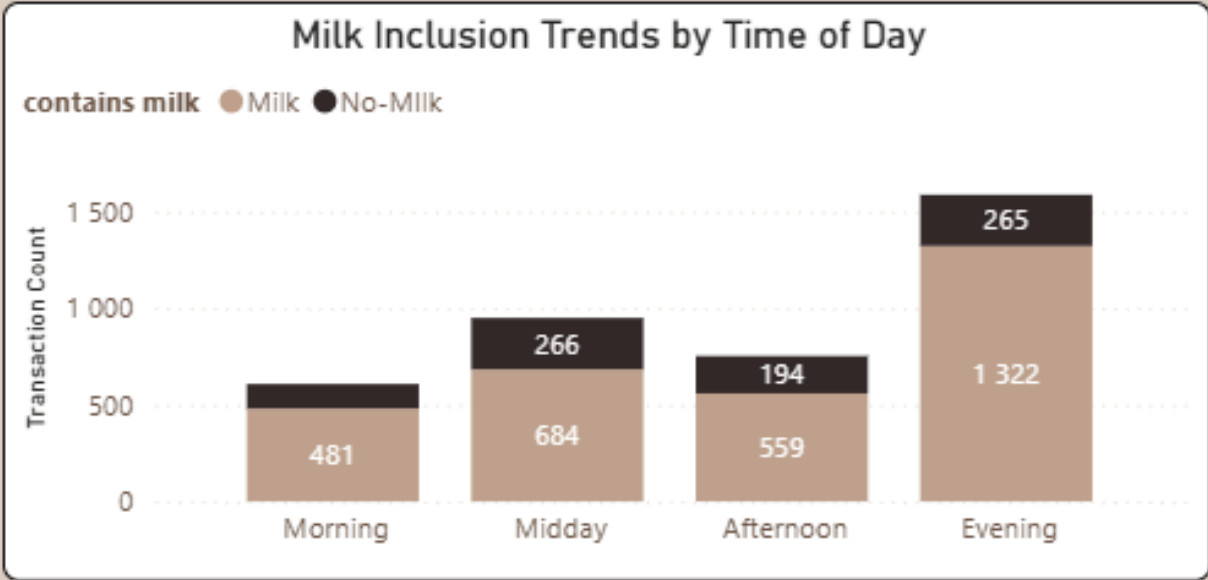
- **Insight:** Milk-based beverages represent the absolute anchor of the shop's product mix, driving the majority of standard volume.
- **Recommendation:** Execute a menu premiumization strategy (introducing craft syrups and organic dairy alternatives) to up-sell within the core product family and drive up the Average Ticket Size.



EXECUTIVE SALES INSIGHTS



Operational Insights

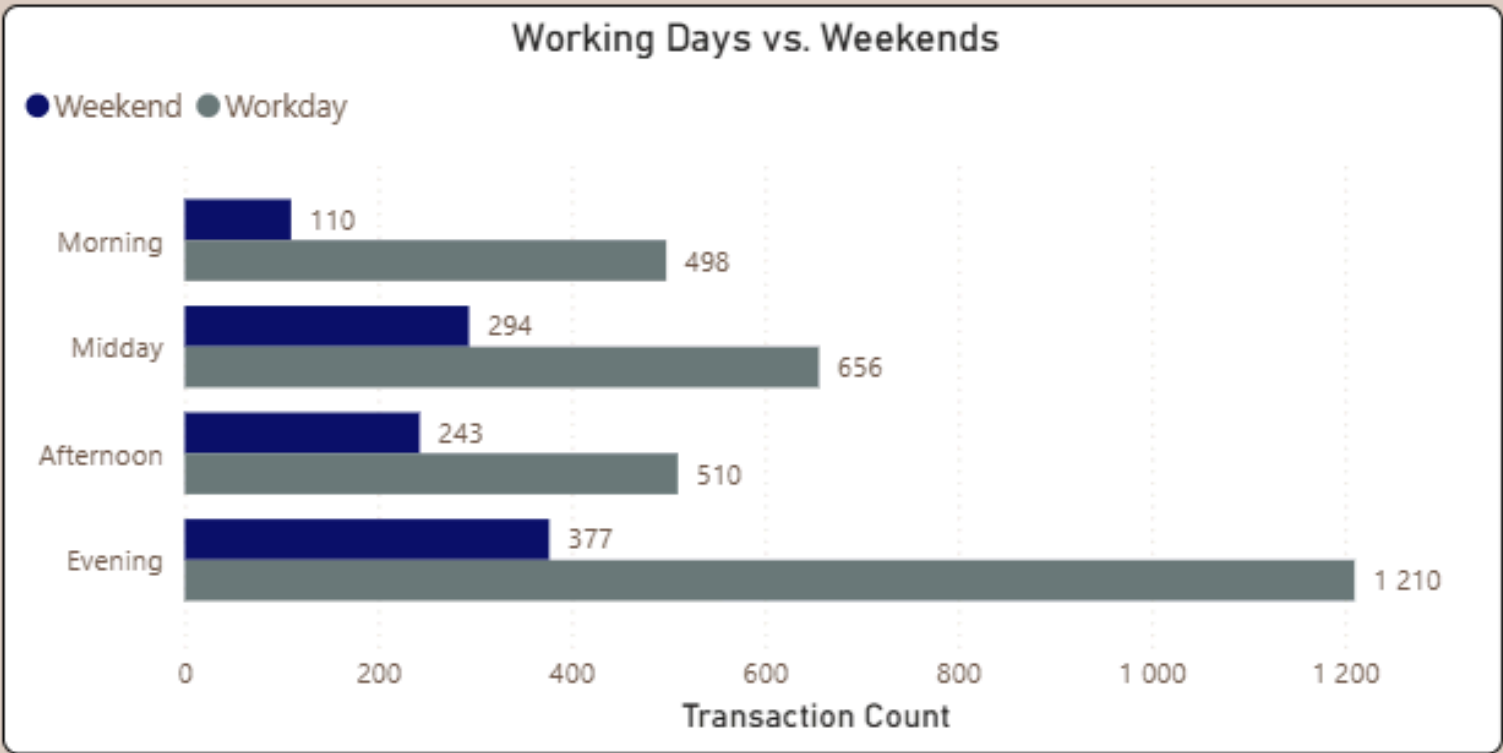


Caffeine Demand Matrix by Service Window

caffeine_level	Morning	Midday	Afternoon	Evening
None	53	101	97	369
Low	1	7	6	7
Medium	262	359	316	687
High	291	481	333	524
Very High	1	2	1	

Top Performers

coffee_name	Total Revenue	Avg Ticket Size
Latte	★ \$28 079	\$34,84
Americano with milk	★ \$25 669	\$29,57
Cappuccino	★ \$18 146	\$35,10
Americano	☆ \$14 994	\$25,28
Hot Chocolate	☆ \$9 964	\$35,33
Cocoa	☆ \$8 496	\$34,96
Cortado	☆ \$7 347	\$25,16
Espresso	☆ \$3 179	\$20,91
Irish whiskey	☆ \$525	\$25,00
Coffee with Irish whiskey	☆ \$464	\$29,00
Irish whiskey with milk	☆ \$405	\$27,00
Double Irish whiskey	☆ \$336	\$28,00
Chocolate with milk	☆ \$324	\$27,00



EXECUTIVE SALES INSIGHTS

Late-Day Transaction Surge

- **Insight:** Timeline analysis reveals a massive, unconventional spike in customer volume during the Evening window, reaching 1,210 transactions on workdays and 377 on weekends.
- **Recommendation:** Extend full barista staffing into the evening hours instead of cutting labor late in the day, directly protecting the shop's highest-volume revenue window.



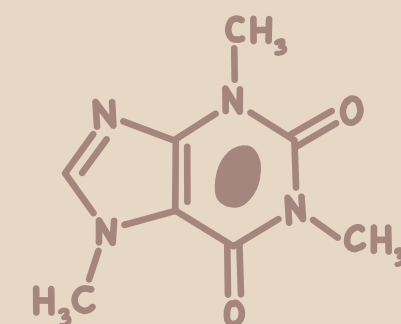
Supply Chain & Milk Inventory Management

- **Insight:** The Milk Inclusion Trends visual proves that milk-based beverages heavily outperform black options across all service windows, peaking at 1,322 transactions in the evening.
- **Recommendation:** Enforce a mandatory dairy and plant-based milk restock right before the afternoon shift change to prevent critical stockouts during peak operational loads.



Evening Caffeine Bifurcation

- **Insight:** The caffeine matrix shows that evening customers split into two extremes: strong Medium/High caffeine drinks (1,211 total) and decaf None options (369).
- **Recommendation:** Optimize evening shift inventory by securing heavy supplies for both traditional high-yield blends and premium non-caffeinated selections.



EXECUTIVE SUMMARY

- **Revenue Growth (Strategic Menu Engineering):** Identified a critical summer assortment gap (lack of iced beverage innovations) and pinpointed key premium margin drivers (the Latte family). This enables targeted up-selling strategies and structural menu optimization.
- **Operational Efficiency (Data-Driven Staffing):** Translated granular customer traffic patterns (weekday morning rushes and weekend evening anomalies) into dynamic shift-scheduling templates. This directly reduces labor costs while eliminating customer wait times.
- **Supply Chain Risk Mitigation (Preventative Inventory Modeling):** Leveraged raw material consumption rates (dairy, plant-based alternatives, and decaf coffee) to establish robust PAR levels. This proactive modeling mitigates the risk of revenue-losing stockouts.

